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Beja Higher School of Technology and Management

Computer Engineering Course

Human-Computer Interaction

Group Work - Part 1

"Analysis and Design of an Interactive System"

Draft Version

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Analysis and Design of an Interactive System

In this group project, students are expected to design the interface of a
Android application. Some examples of working themes are provided, but the
Students may suggest other topics of interest to them, provided they are familiar with the subject matter.
the topic to be addressed, and that they can contact the end users. For the interface design of
In this application, students should refer to the principles and techniques covered in class.
Human-Computer Interaction. Students only need to draw a part.
of the application interface. The part of the interface to be developed should be related
with the application's main tasks.

The project team will consist of two or three members. In choosing the project
Students must ensure that there are enough user tasks to distribute.
by the various elements of the group. In situations where there may not be tasks of
Solutions can be explored for all members of the group.
There are alternatives for the same task, but these alternatives must make sense.

The distribution of project tasks among team members should ensure that
All students participate in all tasks. For example, in the characterization of
For users, each team member must create a user profile, or persona.
In characterizing similar systems, each team member must
to characterize a similar system, and so on. In the end, the work should be
consistent as a whole, but each student will be responsible for the tasks that
developed in the project. Each student must submit a report detailing the...
Tasks completed. In the project presentation, each student must also present...
the tasks he performed.

Students will need to follow a Scrum methodology for project development.
To support project management, students should choose the tool that...
Consider the following as more suitable: Trello, Jira, GitLab, Monday, Notion, ClickUp. The management of
The project should be structured using Epics, Tasks, and Sprints. When creating the team...

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Work with these tools should include the instructor of this course unit.
(using the email luisbgarcia@ipbeja.pt) so that he can follow the work. The
Students should also create a repository on GitHub. to manage the various versions
of the project documents, and share this repository with the unit's lecturer.
The non-functional prototype of the application will be developed using Balsamiq.
Students should follow online tutorials to learn how to use these.
tools.

To receive *feedback* on their work, students must complete the following:
The tasks listed below should be performed gradually.

Tools and Team Organization

(2 points)

1. In this task, students must choose the project theme and explore the...
Proposed work tools, and organizing teamwork. Management
The project should be structured using Epics, Tasks, and Sprints. Each student will have...
the responsibility of fulfilling the set of tasks assigned to them.
In their final individual report, each student must indicate how the project was organized.
teamwork, submitting records that prove your contribution.
for the project and to reflect on the functioning of the work team. **The**
The study of tools and work organization should be carried out in
team.

Participation in Computer Science Events at IPBeja

(1 point)

2. Students are expected to participate in engineering-related events.
computer science events taking place at IPBeja. With participation in these events
The aim is for students to gain a better understanding of how things work.
companies, how they work as teams, the projects developed, the type of
tools and technologies used. Students should apply these.
knowledge in the development of this project. To obtain this quote

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Students must sign the attendance sheet for the events indicated by...

The following events are planned for the teaching staff:

- Computer Science Day

  Several speakers

  Organized by the Scientific and Pedagogical Committee of LEI

  April 8th at 2:30 PM, Auditorium 1

- AI in software engineering

  Filipe Cabaço, Software Engineer at Supabase

  Organized by the Google Developer Group of IPBeja

  April 15th at 11:30 AM, Auditorium 1

Similar Systems

(2 points)

3. Study of systems similar to, or related to, the application that...

The team is developing. At least 2 or 3 should be referenced.

Applications, depending on the number of elements in the group. They should be

Positive and negative aspects of these applications' interfaces are indicated. **Each**

The student must perform an analysis of a similar or related system.

User Characterization

(2 points)

4. Characterization of users based on internet searches or use of...

other sources of information. For example, among other data, the students

They can identify age, gender, education level, type of courses,

computer training or user experience with another type of

technology or computer applications. After this initial characterization, the

Students should think about how they would obtain more information if it were possible.

to contact the users themselves, and the organizations where they work.

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In this stage, students must present 2 or 3 profiles, or personas. representative of the users, depending on the number of elements of group. The credibility of each profile or persona, as a user of the system, Your reason for using the application must be justified by explaining your interest. **Each The student must perform an analysis of a similar or related system.**

Use Cases

(2 points)

5. Development of 2 or 3 application usage scenarios, depending on the number of elements in the group. Only scenarios of the following should be considered: relevant uses for the application, and not secondary scenarios. **Each student You must create a usage scenario for the application.**

Task Analysis

(2 points)

6. Development of hierarchical task analysis for the scenarios described. In the previous step, **each student must perform a hierarchical task analysis. for the use case scenario you studied.**

Style and Interaction Devices

(1 point)

7. Identifying the appropriate interaction style and interaction devices for The chosen application. The options chosen must be justified. **A The identification of the style and interaction devices should be carried out in team.**

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Application Prototype

(6 points)

8. Development of a low-fidelity prototype of the application. The prototype

It should give an overview of how the application works. Think of it this way.

This prototype should be able to describe how the application works to...

The prototype should be organized by user tasks. Each

The student must develop the part of the prototype related to the task of

The user that was assigned to him. Despite this division of labor, the

Students should come together to make decisions that are transversal to the various [issues/issues].

The user's tasks, and thus be able to create a consistent application.

The prototype should consist of the following elements:

(2 points)

a. Sketches of the application screens. The first version of these screens should

to be done in pencil and paper. Later versions may be

developed with a computer support tool for

Screen development. Pay attention to screen readability.

because this aspect will also be valued. It should also be provided.

A brief explanation of how each screen works. **Each student**

must develop screens related to the user's task that

It was distributed to him.

(2 points)

b. Diagrams that illustrate navigation within the application. These diagrams

They can be produced using a tool of the students' choice.

A brief explanation of how the process works should be provided.

Navigation within the application. **Each student must develop the diagrams.**

that illustrate the navigation in the user's task that was assigned to them.

distributed.

(2 points)

c. Indication of how usability principles were applied in

application. The answer to this question should consist of a list with

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the most important usability principles and rules, at most 3 or 4 (not all…), and an explanation of how they were applied in the design. In addition to these principles and rules, students may also indicate how did they apply other knowledge from the area of person-to-person interaction? computer in the application design. **Each student must justify the design your part of the prototype based on these principles and rules.**

Presentation of the Work

(2 points)

9. Each student must prepare a report presenting their contributions to the project. The report should indicate in detail how it was carried out. AI was used in various tasks of the project. The presentation will take place in as a whole, but each team member must describe and present their own. This is part of the work. The presentation should have a maximum duration of 15 minutes. for groups of 2 people and 20 minutes for groups of 3 people. The Spending too much time on the presentation will penalize the group on this point. **The report It should be done individually and presented as a group, but with each member of the group presenting their contributions to the project. Students will be required to present the entire development process. of the prototype and not just the final result.**

Working Groups

The laboratory should be developed by groups of 2 or 3 members. They are not... Changes to the group elements are permitted except in extraordinary situations and duly justified.

Work Submission

Students must complete the specified tasks and submit their solutions in a They must also submit an electronic presentation describing the report.

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Application design. The name of this file must contain the indication TG1, followed by The team number and finally the numbers of the group members. Each of these Elements must be separated by an *underscore* (_). For example TG1_03_1000_2000_3000.zip would be the Group Work 1 of team 3 composed of Students with the numbers 1000, 2000, and 3000. The assignment must be submitted on completed through the course's *Moodle* page. **The following will not be considered. Resolutions delivered by email.**

Assessment

Only work partially completed in class will be accepted. Solutions developed exclusively outside of class are not accepted. The various Team members should participate in the preparation of the work. The records The actions taken using the various work tools should confirm that the work was Performed by all team members and distributed over time. No. Copying will be tolerated in the submitted solutions. In these situations, students will obtain Zero grade on the assignment. When necessary, a discussion can be scheduled with the... students. Any member of the group should be able to answer the questions. placed.

Good job

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